



BIT4139 Ethical Hacking and Counter Measures

BSCCS/2024/62210 EVANS MUTHOMI KABURU

Tool of choice-Nmap

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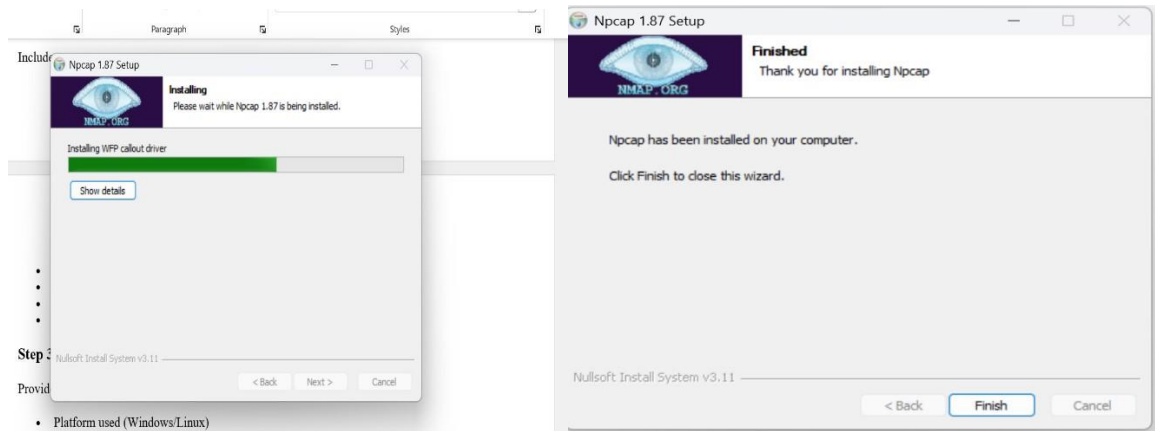
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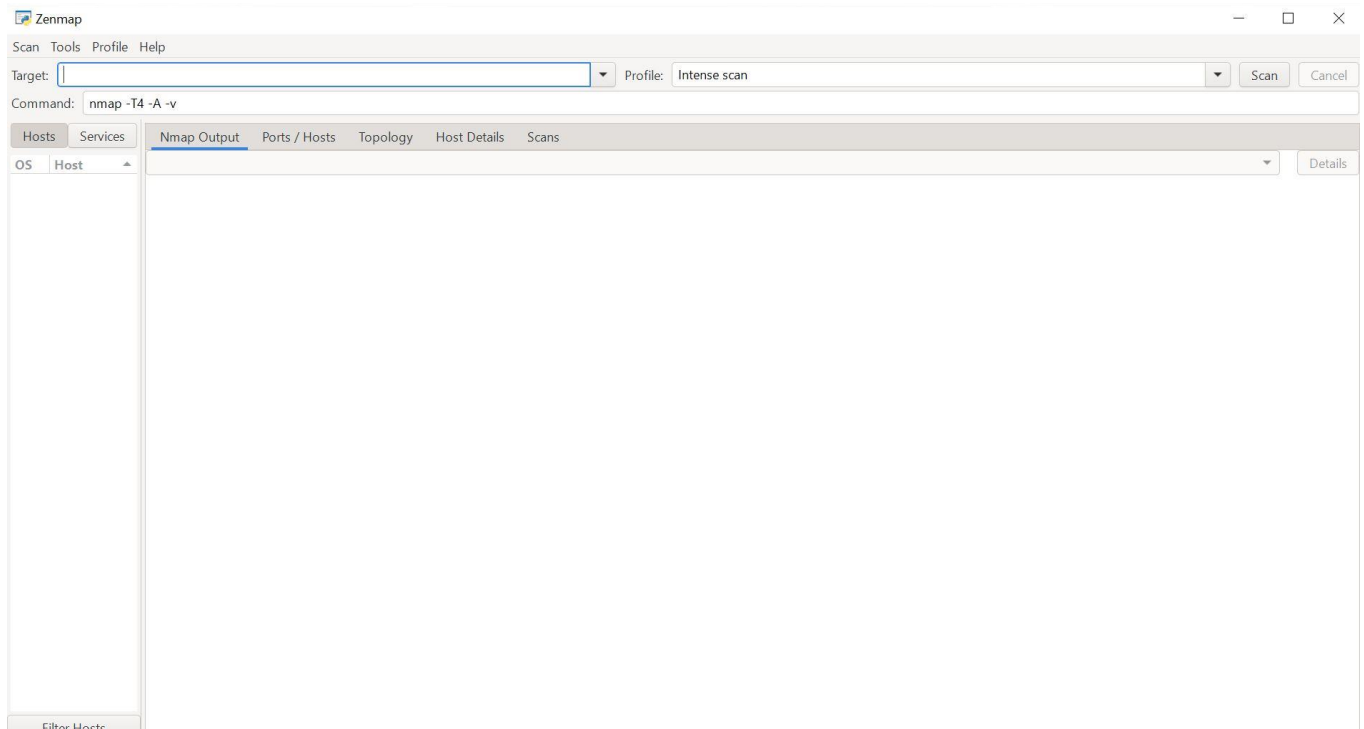
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Week 1-3 work

Installing of Nmap



User Interface



Version check

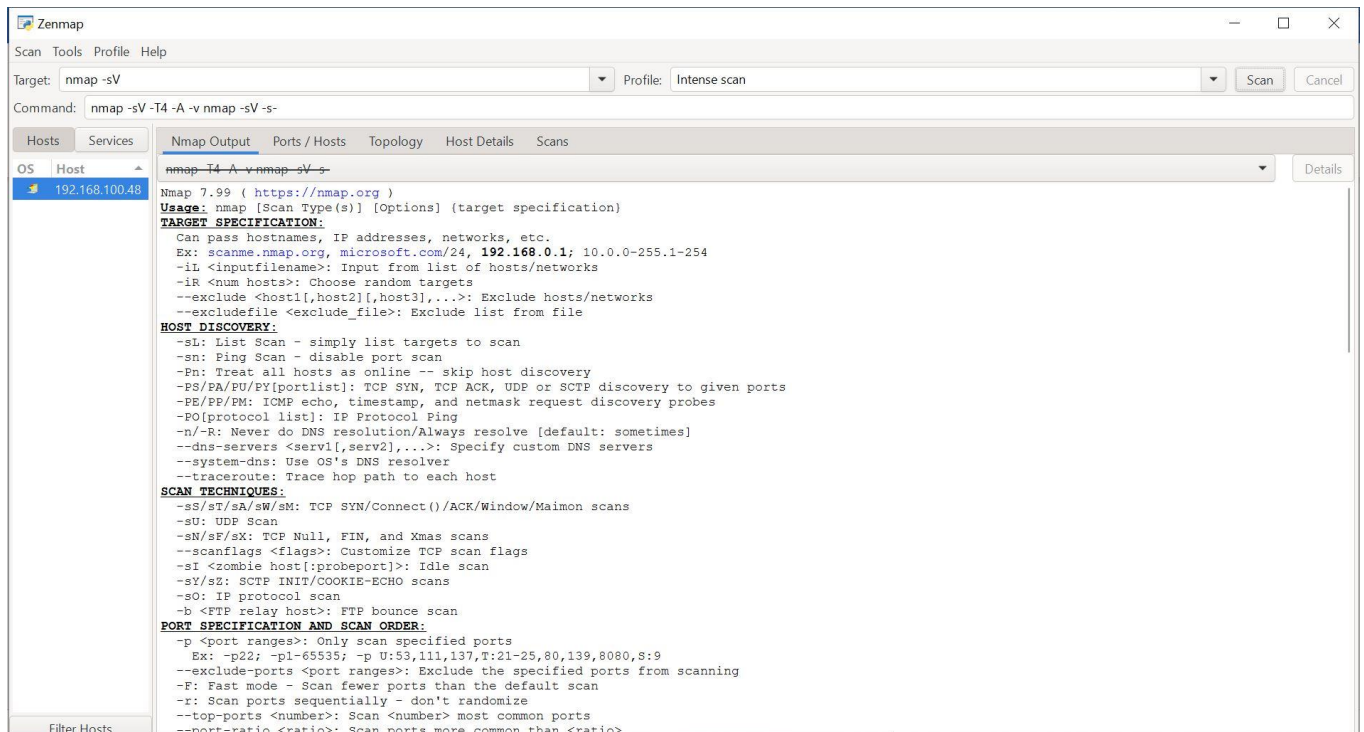
```
Command Prompt
Microsoft Windows [Version 10.0.19045.7291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ADMIN>NMAP -SV
Failed to resolve/decode supposed IPv4 source address "V": No such host is known.
QUITTING!

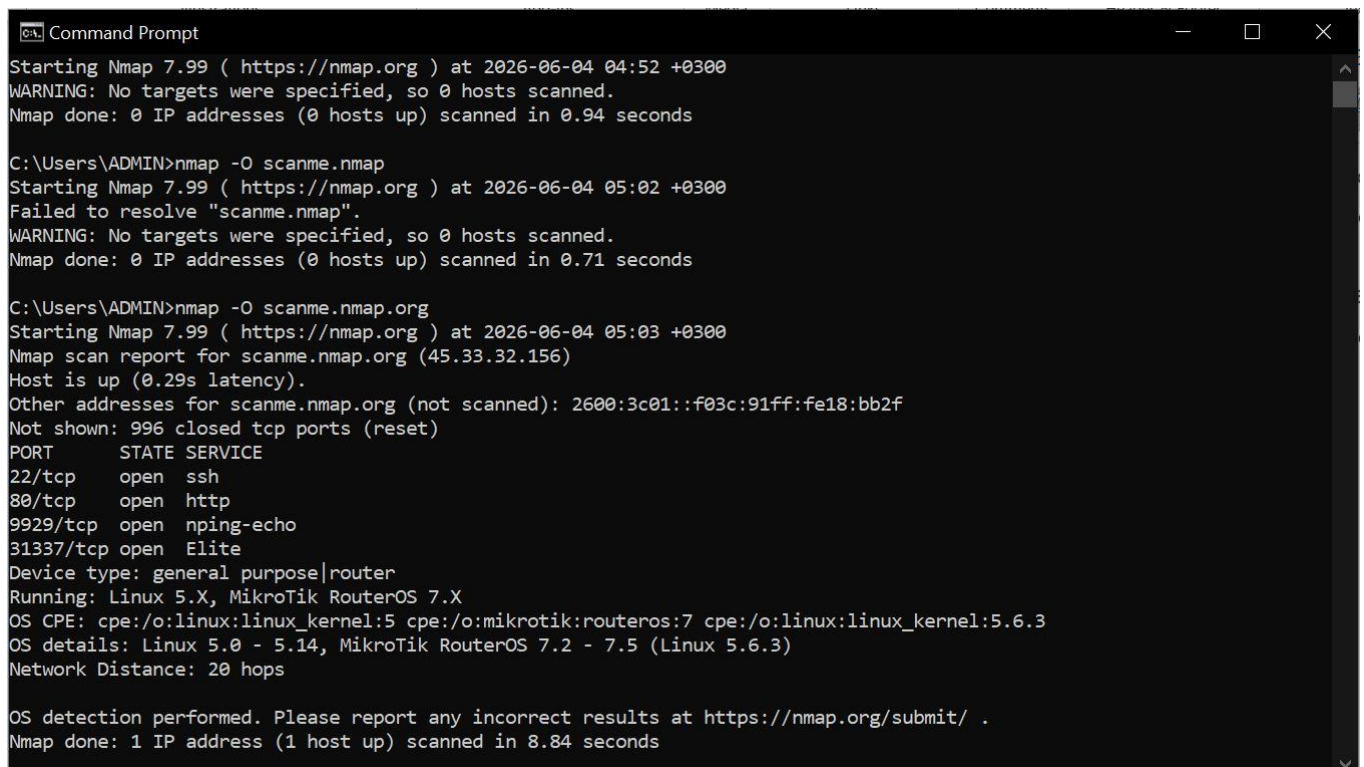
C:\Users\ADMIN>nmap -sV
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 04:52 +0300
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.94 seconds

C:\Users\ADMIN>
```

Version Check on the UI



OS scan



OS scan on the GUI

The screenshot shows the Zenmap application window. At the top, the 'Target' field contains 'nmap -O scanme.nmap.org' and the 'Profile' is set to 'Quick scan'. The 'Command' field shows the full Nmap command: 'nmap -T4 -O -F nmap -O scanme.nmap.org -O -O -O -O -O -O -O -O'. The main panel displays the 'Nmap Output' for the scan of 'scanme.nmap.org' (45.33.32.156). The output includes the start time, host status, open ports (ssh, http), device type (general purpose router), running OS (Linux 5.X), OS CPE, OS details, network distance, and scan completion information.

Starting Nmap 7.99 (<https://nmap.org>) at 2026-06-04 05:09 +0300
Failed to resolve "nmap".
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 98 closed tcp ports (reset)
PORT STATE SERVICE
22/tcp open ssh
80/tcp open http
Device type: general purpose|router
Running: Linux 5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 5.0 - 5.14, MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 20 hops

OS detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 6.96 seconds

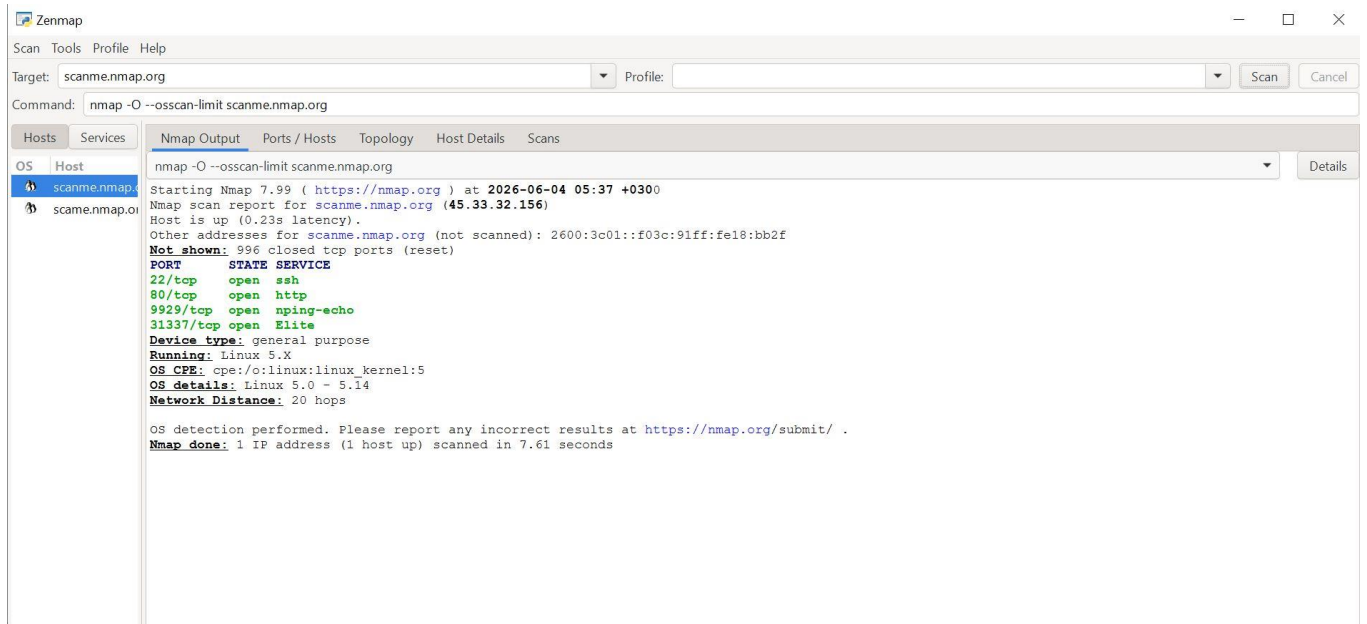
Narrowed down OS Scan

```
Command Prompt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 05:05 +0300
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.17 seconds

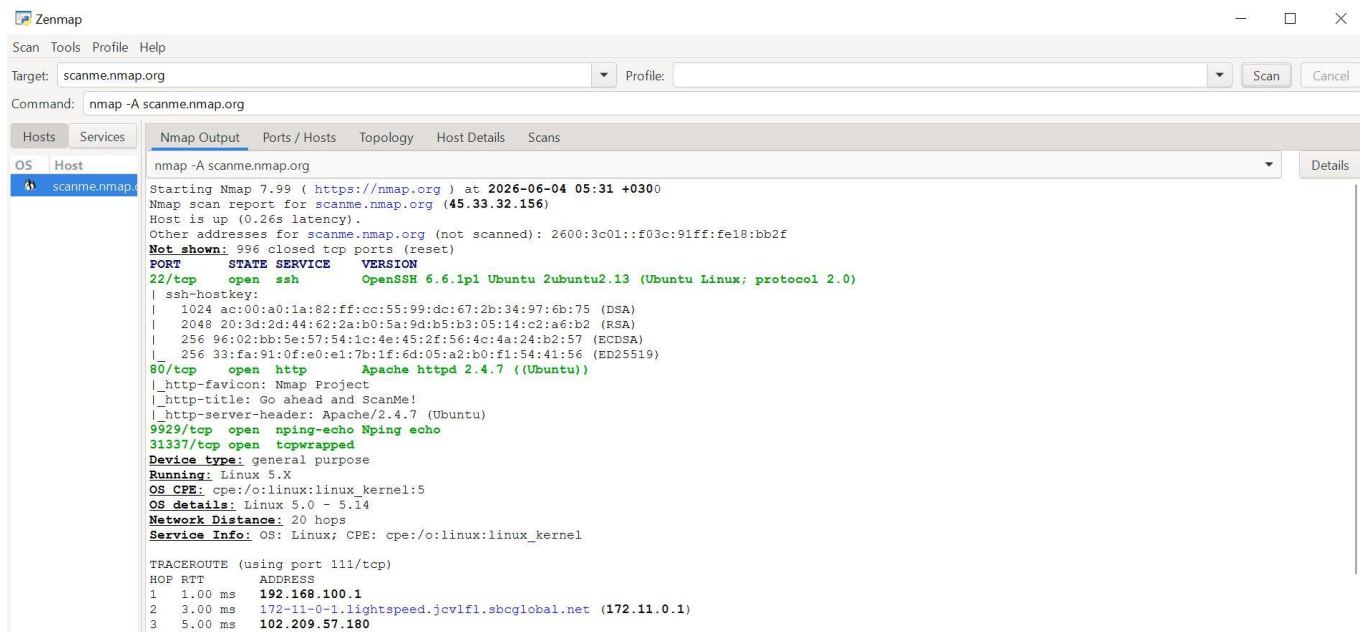
C:\Users\ADMIN>nmap -O --osscan-limit scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 05:15 +0300
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite
Device type: general purpose
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.14
Network Distance: 20 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.18 seconds

C:\Users\ADMIN>nmap -O --osscan-limit scanme.nmap.org
```



Aggressive Scans



The screenshot shows the Zenmap application window. The 'Hosts' tab is selected, displaying a list of hosts with 'scanme.nmap.org' highlighted. The 'Nmap Output' tab is also visible, showing the results of an aggressive scan (-A) performed on scanme.nmap.org at 2026-06-04 05:31 +0300. The output includes details about the host's IP (45.33.32.156), latency (0.26s), and various open ports and services.

Hosts Services

OS Host

scanme.nmap.org

Command: nmap -A scanme.nmap.org

Target: scanme.nmap.org Profile:

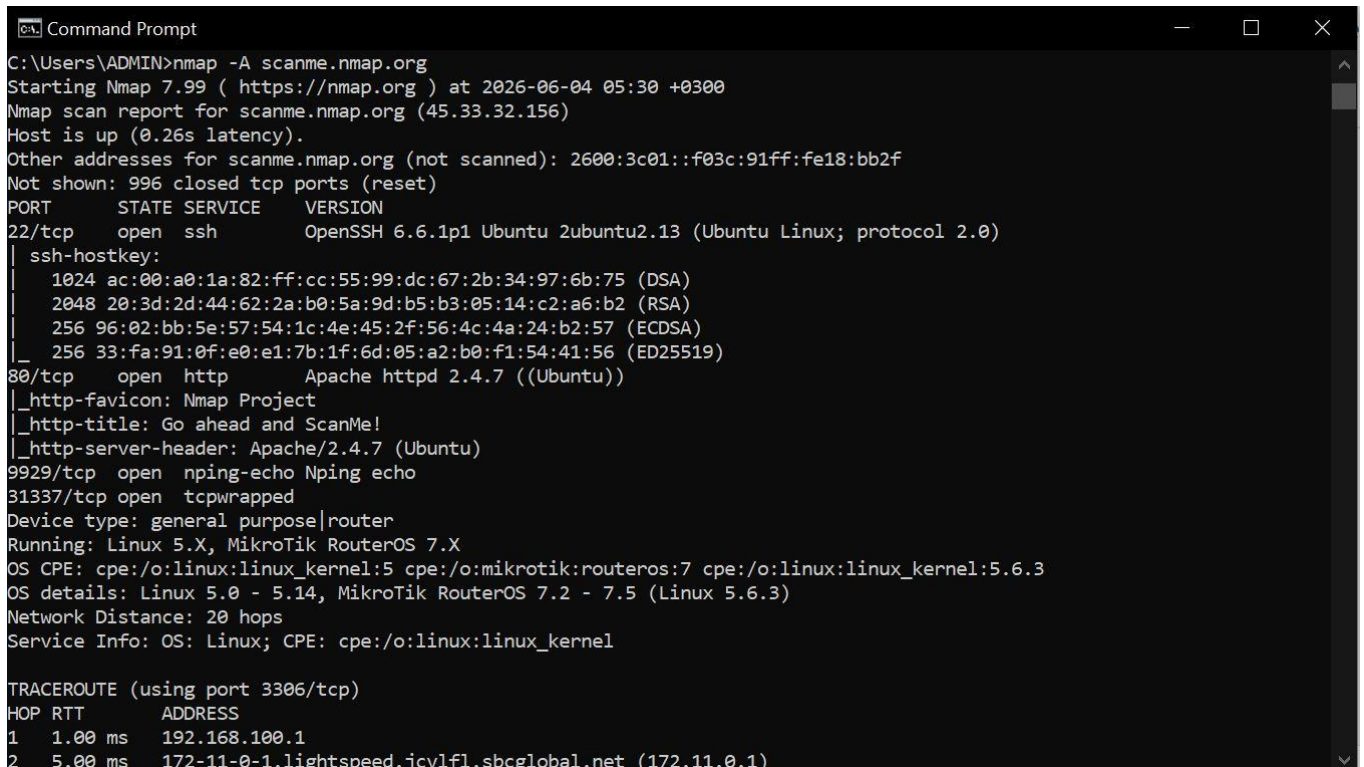
Scan Cancel

Nmap Output Ports / Hosts Topology Host Details Scans

Starting Nmap 7.99 (https://nmap.org) at 2026-06-04 05:31 +0300
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
|_ http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp open nping-echo Nping echo
31337/tcp open tcpwrapped
Device type: general purpose
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.14
Network Distance: 20 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 111/tcp)
HOP RTT ADDRESS
1 1.00 ms 192.168.100.1
2 3.00 ms 172-11-0-1.lightspeed.jcvlfl.sbcglobal.net (172.11.0.1)
3 5.00 ms 102.209.57.180

Aggressive scan on GUI



The screenshot shows a Windows Command Prompt window with the output of an aggressive scan (-A) performed on scanme.nmap.org at 2026-06-04 05:30 +0300. The output is identical to the one shown in the Zenmap GUI, including details about the host's IP (45.33.32.156), latency (0.26s), and various open ports and services.

Command Prompt

C:\Users\ADMIN>nmap -A scanme.nmap.org
Starting Nmap 7.99 (https://nmap.org) at 2026-06-04 05:30 +0300
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
|_ http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp open nping-echo Nping echo
31337/tcp open tcpwrapped
Device type: general purpose|router
Running: Linux 5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 5.0 - 5.14, MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 20 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 3306/tcp)
HOP RTT ADDRESS
1 1.00 ms 192.168.100.1
2 5.00 ms 172-11-0-1.lightspeed.jcvlfl.sbcglobal.net (172.11.0.1)

Host scanning

```
Command Prompt

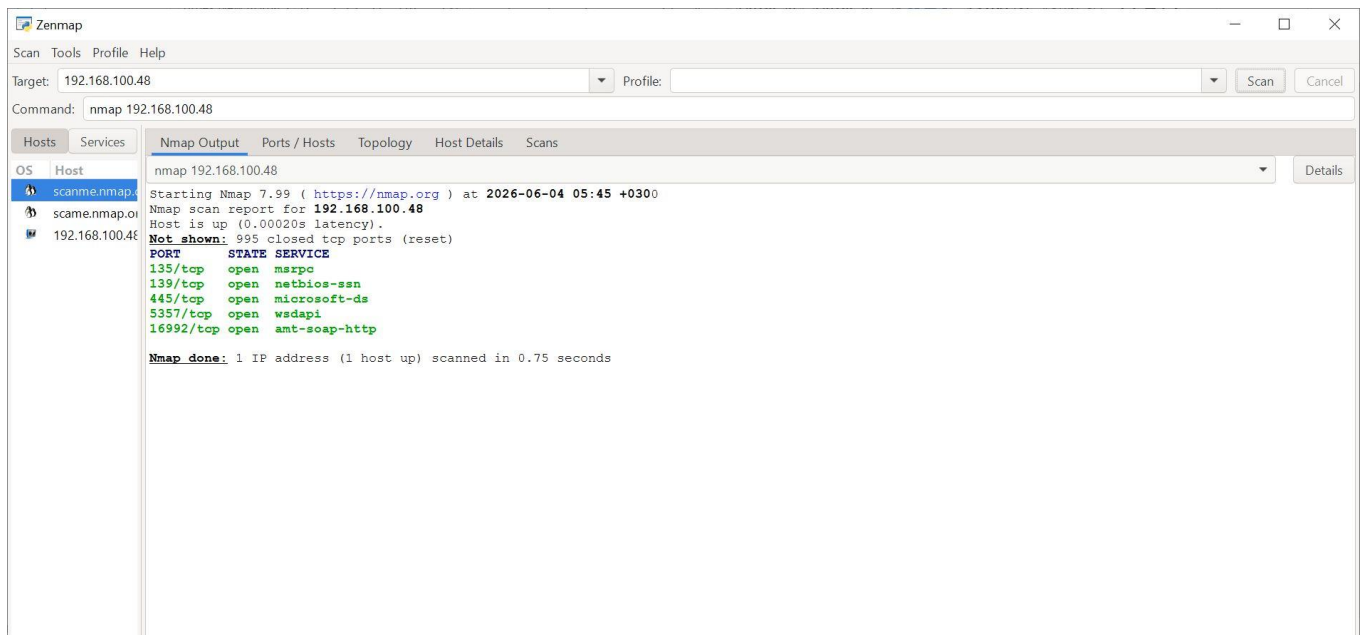
'$nmap' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ADMIN>$ nmap 192.168.100.48
'$' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ADMIN>nmap 192.168.100.48
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 05:44 +0300
Nmap scan report for 192.168.100.48
Host is up (0.000042s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsddapi
16992/tcp  open  amt-soap-http

Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds

C:\Users\ADMIN>
```



Scanning Multiple hosts

The screenshot shows the Zenmap application window. The 'Target' field is set to '192.168.100.1-100' and the 'Command' is 'nmap 192.168.100.1-100'. The 'Hosts' tab is active, displaying a list of hosts: 192.168.100.1, 192.168.100.44, and 192.168.100.48. The 'Nmap Output' tab is selected, showing the scan results for these hosts. The output includes the start time, host status, and a table of open ports and services for each host.

```
nmap 192.168.100.1-100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 05:49 +0300
Nmap scan report for 192.168.100.1
Host is up (0.0037s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    open  telnet
53/tcp    open  domain
80/tcp    open  http
MAC Address: 00:11:41:12:09:E8 (GoodMan)

Nmap scan report for 192.168.100.44
Host is up (0.0069s latency).
All 1000 scanned ports on 192.168.100.44 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 4E:9F:38:C7:01:A6 (Unknown)

Nmap scan report for 192.168.100.48
Host is up (0.00059s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsdapi
16992/tcp open  amt-soap-http

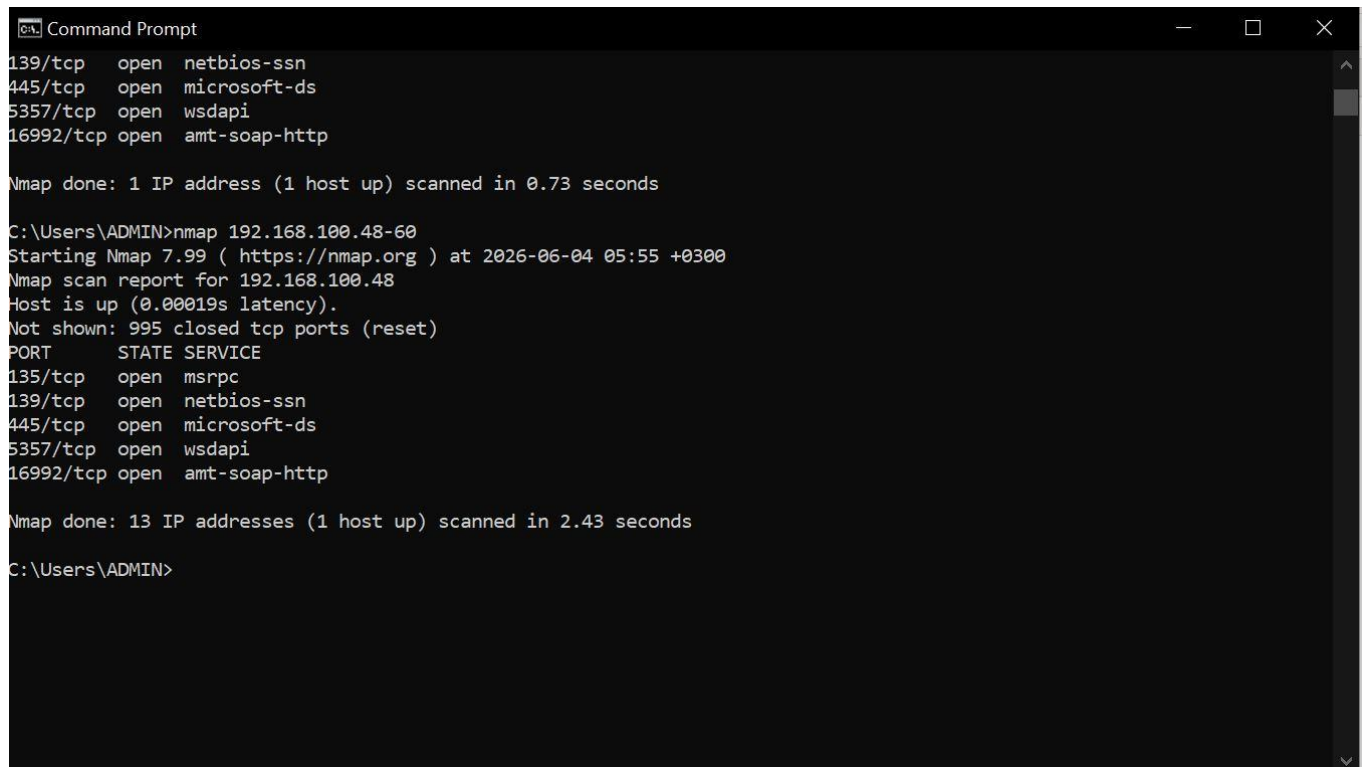
Nmap done: 100 IP addresses (3 hosts up) scanned in 7.91 seconds
```

The screenshot shows the Zenmap application window. The 'Target' field is set to '192.168.100.48,49,45,46' and the 'Command' is 'nmap 192.168.100.48,49,45,46'. The 'Hosts' tab is active, displaying a list of hosts: 192.168.100.1, 192.168.100.44, and 192.168.100.48. The 'Nmap Output' tab is selected, showing the scan results for these hosts. The output includes the start time, host status, and a table of open ports and services for each host.

```
nmap 192.168.100.48,49,45,46
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 05:52 +0300
Nmap scan report for 192.168.100.48
Host is up (0.0000070s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsdapi
16992/tcp open  amt-soap-http

Nmap done: 4 IP addresses (1 host up) scanned in 2.24 seconds
```

Scanning multiple hosts on CMD



```
139/tcp open netbios-ssn
445/tcp open microsoft-ds
5357/tcp open wsddapi
16992/tcp open amt-soap-http

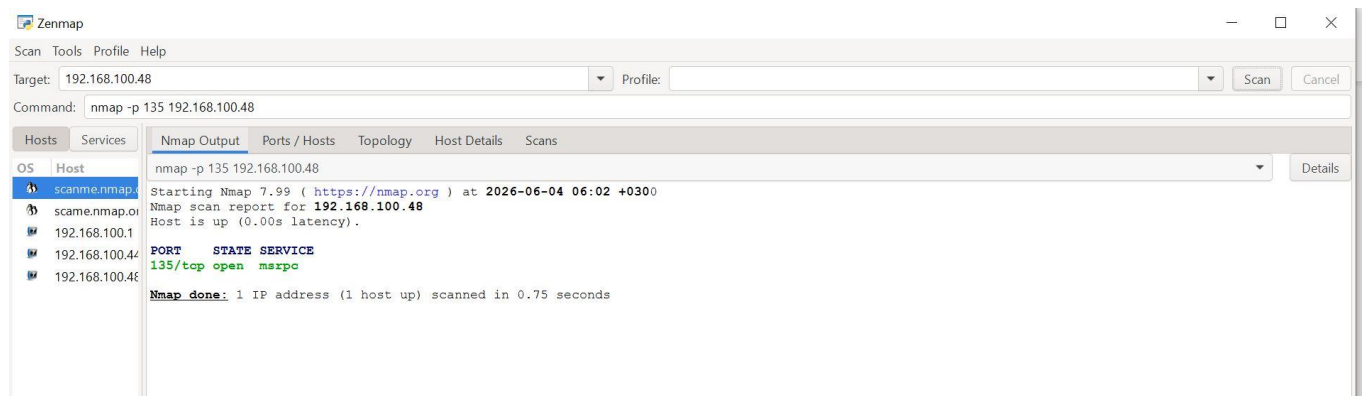
Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds

C:\Users\ADMIN>nmap 192.168.100.48-60
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 05:55 +0300
Nmap scan report for 192.168.100.48
Host is up (0.00019s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsddapi
16992/tcp open  amt-soap-http

Nmap done: 13 IP addresses (1 host up) scanned in 2.43 seconds

C:\Users\ADMIN>
```

Port scanning



Port scan on CMD

```
Command Prompt
C:\Users\ADMIN>nmap -p 988 192.168.100.48
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 06:04 +0300
Nmap scan report for 192.168.100.48
Host is up (0.0010s latency).

PORT      STATE SERVICE
988/tcp   closed unknown

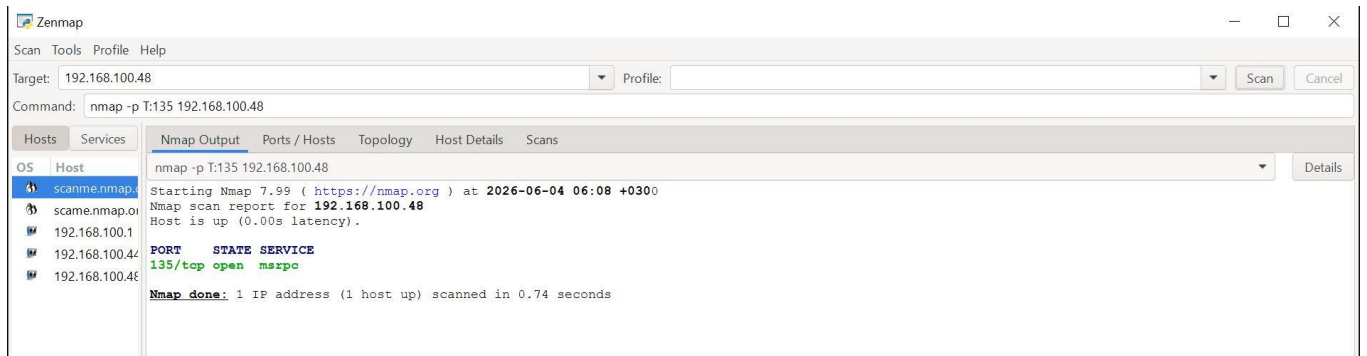
Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds

C:\Users\ADMIN>nmap -p 135 192.168.100.48
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 06:05 +0300
Nmap scan report for 192.168.100.48
Host is up (0.0010s latency).

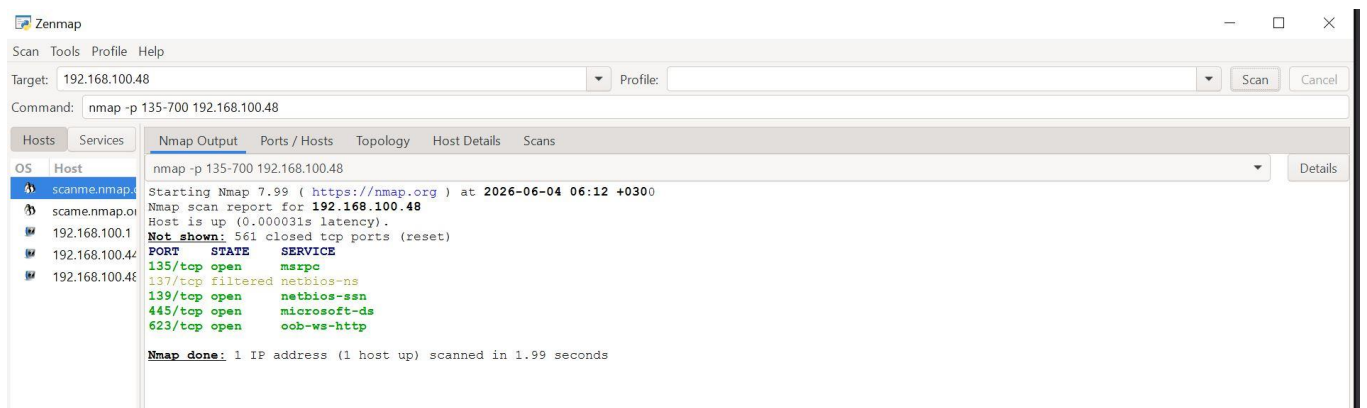
PORT      STATE SERVICE
135/tcp   open  msrpc

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

TCP port scan



Scanning a range of TCP ports



Scanning top 5 ports in my DNS

The screenshot displays the Zenmap application window. At the top, the menu bar includes 'Scan', 'Tools', 'Profile', and 'Help'. The 'Target' field is set to '192.168.100.1', and the 'Command' field contains 'nmap --top-ports 5 192.168.100.1'. The 'Scan' button is visible next to the command field.

The main window is divided into several tabs: 'Hosts', 'Services', 'Nmap Output', 'Ports / Hosts', 'Topology', 'Host Details', and 'Scans'. The 'Nmap Output' tab is currently selected, showing the following text:

```
nmap --top-ports 5 192.168.100.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-04 06:16 +0300
Nmap scan report for 192.168.100.1
Host is up (0.0031s latency).

PORT      STATE      SERVICE
21/tcp    filtered   ftp
22/tcp    filtered   ssh
23/tcp    open       telnet
80/tcp    open       http
443/tcp   closed     https
MAC Address: 00:11:41:12:09:E8 (GoodMan)

Nmap done: 1 IP address (1 host up) scanned in 2.02 seconds
```

On the left side, under the 'Hosts' tab, a list of hosts is shown. The first host, '192.168.100.1', is highlighted, and its details are visible in the 'Host Details' tab.

Week 4 and 5

Task 1

Sample institution- JPMorgan Chase

Domain info(*chase.com*)

 Domains Hosting Websites Email Security Whois Deals

chase.com

Updated 5 days ago

 **Domain Information**

Domain:	chase.com
Registered On:	1995-10-11
Expires On:	2027-10-10
Updated On:	2025-09-08
Status:	client delete prohibited client transfer prohibited client update prohibited server delete prohibited server transfer prohibited server update prohibited
Name Servers:	ns0119.secondary.cloudflare.com ns0140.secondary.cloudflare.com ns05.jpmorganchase.com ns06.jpmorganchase.com

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Applications

Domains

Hostnames

IPs

Platform Stack

Hosting

SaaS

CDNs

Network Stack

ISPs/Telcos

Special Networks

Other Networks

JPMorgan Chase is a multinational bank that provides credit cards, mortgages, commercial banking, auto loans, investing and retirement planning, checking and business banking.

JPMorgan Chase Domains

Primary Domains

• chase.com

• chasecdn.com

• jpmchase.com

• gslbjpmchase.com

• jpmorgan.com

• jpmorganchase.com

• chasepaymentech.com

• and 13 more

Primary Domains are owned and managed by the application vendor.

Platform Domains are often provided by [hosting providers](#), [content delivery networks](#), [SaaS platforms](#) and other third parties. It is important to identify these domains for DNS filters and hostname blocklists.

• jpmorganchase.com

• chasepaymentech.com

• and 13 more

Platform Domains

Platform	Sample Domain	Details
Adobe Ads	{hidden}.omtrdc.net	
Adobe Campaign	{hidden}.cname.campaign.adobe.com	
Adobe EM	{hidden}.demdex.net	
Adobe Marketo	{hidden}.mktoweb.com	and 1 more
Akamai	{hidden}.akadns.net	and 20 more
LexisNexis Risk	{hidden}.online-metrix.net	
Yext	{hidden}.pagescdn.com	and 1 more

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Core Networks

Applications are sometimes deployed on partner networks, parent company networks, or regional hosting providers. The following is a sample of JPMorgan Chase IP addresses deployed on this type of network.

Network	IP	PoP	Category	Location
RETN	2a02:2d8:0:9181::1ce	Kyiv	ISP/Telco	Ukraine
RETN	2a02:2d8:0:9181::195	Kyiv	ISP/Telco	Ukraine
RETN	2a02:2d8:0:9181::147	Kyiv	ISP/Telco	Ukraine
RETN	2a02:2d8:0:9181::153	Kyiv	ISP/Telco	Ukraine
RETN	2a02:2d8:0:9181::145	Kyiv	ISP/Telco	Ukraine
... and 100 more				

18

Domain registrar (*chase.com*)

 **Registrant Contact**

Name:	Domain Administrator
Organization:	JPMorgan Chase & Co.
Street:	201 North Walnut Street Mail Suite DE1-0175
City:	Wilmington
State:	DE
Postal Code:	19801
Country:	US
Phone:	+1.3022823003
Email:	https://domains.markmonitor.com/whois/contact/chase.com

IP address (*chase.com*)

target: chase.com

Profile: Quick scan

Command: nmap -T4 -F chase.com

Hosts

Services

OS

Host

scanme.nmap.org

scanme.nmap.org

chase.com (146.143.83.57)

chase.com (146.143.141.57)

chase.com (146.143.13.57)

example.com (192.168.100.1)

192.168.100.42

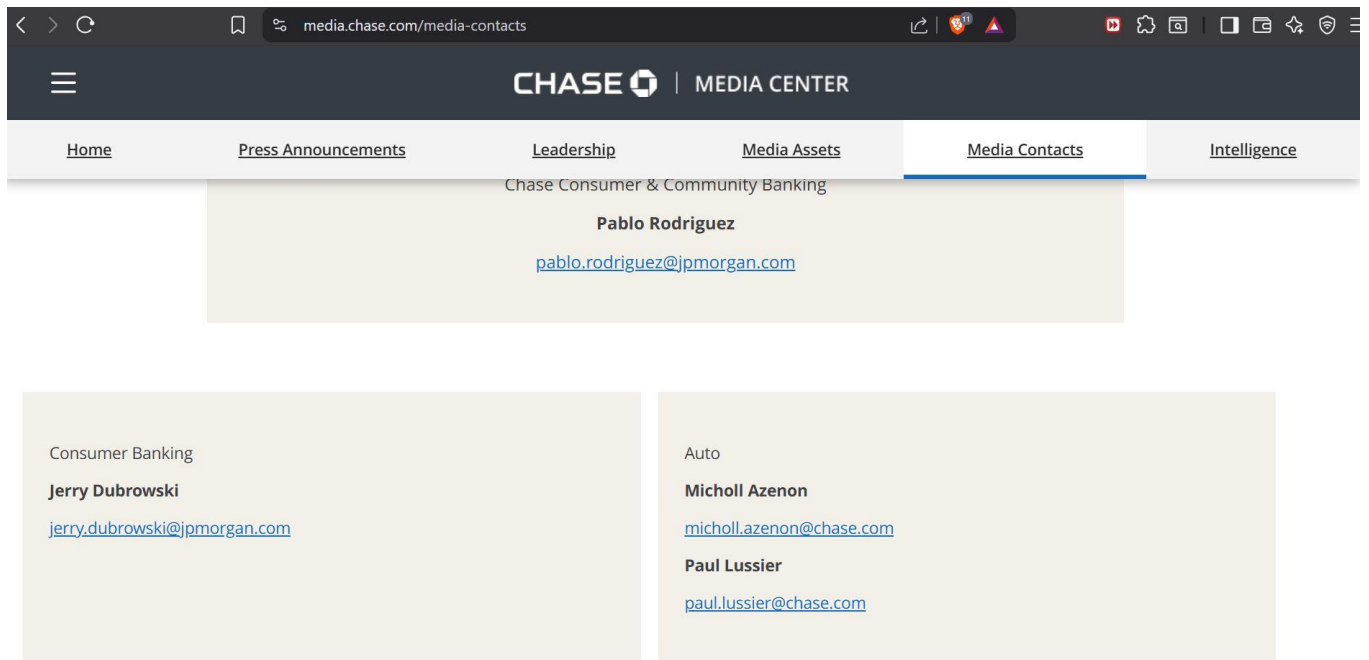
192.168.100.43

nmap -T4 -F chase.com

Starting Nmap 7.99 (<https://nmap.org>) at 2026-06-04 13:43 +0300
Nmap scan report for chase.com (146.143.83.57)
Host is up (0.051s latency).
Other addresses for chase.com (not scanned): 64:ff9b::928f:d39 64:ff9b::928f:8d39 64:ff9b::928f:5339 146.143.141.57 146.143.13.57

PORT	STATE	SERVICE
7/tcp	open	echo
9/tcp	open	discard
13/tcp	open	daytime
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
25/tcp	open	smtp
26/tcp	open	rsftp
37/tcp	open	time
53/tcp	filtered	domain

Email patterns



Having tried multiple scans on JP Morgan Chase website, I realized that I can't really get much from it, with the tools I have on my computer.

Therefore, I switched to *pentest-ground.com* for the sake of completing my class work.

Task 2

Port scanning (*chase.com*)

OS

Host

scanme.nmap.org

scanme.nmap.org

chase.com (146.143.83.57)

chase.com (146.143.83.57)

chase.com (146.143.83.57)

example.com (192.168.100.1)

192.168.100.1

192.168.100.46

192.168.100.46

nmap -T4 -f chase.com

Starting Nmap 7.99 (<https://nmap.org>) at 2026-06-04 13:43 +0300

Nmap scan report for chase.com (146.143.83.57)

Host is up (0.051s latency).

Other addresses for chase.com (not scanned): 64:ff9b::928f:d39 64:ff9b::928f:8d39 64:ff9b::928f:5339 146.143.141.57 146.143.13.57

PORT	STATE	SERVICE
7/tcp	open	echo
9/tcp	open	discard
13/tcp	open	daytime
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
25/tcp	open	smtp
26/tcp	open	rsftp
37/tcp	open	time
53/tcp	filtered	domain
79/tcp	open	finger
80/tcp	open	http
81/tcp	open	hosts2-ns
88/tcp	open	kerberos-sec
106/tcp	open	pop3pw
110/tcp	open	pop3
111/tcp	open	rpcbind
113/tcp	open	ident
119/tcp	open	nnntp
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
143/tcp	open	imap
144/tcp	open	news
179/tcp	open	bgp
199/tcp	open	smux
389/tcp	open	ldap

Details

Detection of the running services (*pentest-ground.com*)

Zenmap

—

□

×

Scan

Tools

Profile

Help

Target:

nmap pentest-ground.com

▼

Profile:

▼

Scan

Cancel

Command:

sudo -sV -p 4280 -Pn nmap pentest-ground.com

Hosts

Services

Nmap Output

Ports / Hosts

Topology

Host Details

Scans

Service

afs3-callback

cslistener

hosts2-ns

http

https

ident

redis

secure-mqtt

upnotifyp

sudo -sV -p 4280 -Pn nmap pentest-ground.com

▼

Details

Starting Nmap 7.99 (<https://nmap.org>) at 2026-06-11 10:20 +0300

Failed to resolve "nmap".

Nmap scan report for pentest-ground.com (178.79.134.182)

Host is up (0.18s latency).

rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE	VERSION
4280/tcp	open	ssl/http	nginx 1.31.1

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 18.10 seconds

OS scan (pentest-ground.com)

 Zenmap

Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services

Service

afs3-callback
cslister
hosts2-ns
http
https
ident
redis
secure-mqtt
upnotifyp
vrml-multi-use

Nmap Output Ports / Hosts Topology Host Details Scans

sudo -p 4280 -O -Pn pentest-ground.com

Starting Nmap 7.99 (<https://nmap.org>) at 2026-06-11 12:47 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.13s latency).
Other addresses for pentest-ground.com (not scanned): 64:ff9b::b24f:86b6
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE
4280/tcp	filtered	vrml-multi-use

Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing an open TCP port so results incomplete
No OS matches for host

OS detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 16.09 seconds

Web Technology Enumeration (*pentest-ground.com*)

Zenmap

Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services **Nmap Output** Ports / Hosts Topology Host Details Scans

Service

- afs3-callback
- cslistener
- hosts2-ns
- http
- https
- ident
- redis
- secure-mqtt
- upnotifyp

```
nmap -sV -p 4280 -Pn --script http-vuln* pentest-ground.com

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-11 12:54 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.99s latency).
Other addresses for pentest-ground.com (not scanned): 64:ff9b::b24f:86b6
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT      STATE SERVICE VERSION
4280/tcp  open  ssl/http nginx 1.31.1
|_http-server-header: nginx/1.31.1

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 51.43 seconds
```

Week 6

Host Discovery

Confirms host is live

The screenshot displays the Zenmap application window. At the top, the title bar reads "Zenmap". Below it is a menu bar with "Scan", "Tools", "Profile", and "Help". The main interface is divided into several sections. On the left, there is a sidebar with "Hosts" and "Services" tabs. Under the "Hosts" tab, a list shows "OS" and "Host" columns, with "192.168.100.51" selected. The main area on the right shows the "Nmap Output" tab, which contains the following text: "nmap -sn 192.168.100.51", "Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 05:59 +0300", "Nmap scan report for 192.168.100.51", "Host is up.", and "Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds".

Zenmap

Scan Tools Profile Help

Target: 192.168.100.51 Profile:

Command: nmap -sn 192.168.100.51

Hosts Services

Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

192.168.100.51

nmap -sn 192.168.100.51

Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 05:59 +0300

Nmap scan report for 192.168.100.51

Host is up.

Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds

Port scanning and OS detection

Zenmap

Scan Tools Profile Help

Target: 192.168.100.51 Profile:

Command: `nmap -sS -O 192.168.100.51`

Hosts Services

OS Host

192.168.100.51

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -sS -O 192.168.100.51

Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 06:01 +0300
Nmap scan report for 192.168.100.51
Host is up (0.00025s latency).
Not shown: 995 closed tcp ports (reset)

PORT	STATE	SERVICE
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
5357/tcp	open	wsdapi
16992/tcp	open	amt-soap-http

Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1809 - 21H2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds

Service and Version Detection

Zenmap

Scan Tools Profile Help

Target: 192.168.100.51 Profile:

Command: `nmap -sV 192.168.100.51`

Hosts Services

OS Host

192.168.100.51

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -sV 192.168.100.51

Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 06:02 +0300
Nmap scan report for 192.168.100.51
Host is up (0.00011s latency).
Not shown: 995 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds?	
5357/tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
16992/tcp	open	http	Intel Active Management Technology User Notification Service httpd 11.8.94.4494

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:11.8.94.4494

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 13.76 seconds

Vulnerability Scan with NSE Scripts

*Targeting specific ports since the general scan is blocked by host or firewalls

The screenshot displays the Zenmap application window. At the top, there is a menu bar with 'Scan', 'Tools', 'Profile', and 'Help'. Below the menu, the 'Target' field is set to '192.168.100.51' and the 'Command' field contains 'nmap -sV -p 80,443 --script vuln 192.168.100.51'. The main interface is divided into two panes. The left pane, titled 'Hosts' and 'Services', shows a list of hosts with '192.168.100.51' selected. The right pane, titled 'Nmap Output', displays the scan results. The output text is as follows:

```
nmap -sV -p 80,443 --script vuln 192.168.100.51
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 06:14 +0300
Nmap scan report for 192.168.100.51
Host is up (0.0010s latency).

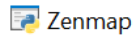
PORT      STATE SERVICE VERSION
80/tcp    closed http
443/tcp   closed https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.11 seconds
```

Week 7

Directory Enumeration Scan

Finds hidden directories/forms-Broken Access Control.



Scan Tools Profile Help

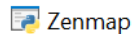
Target: Profile:

Command:

Hosts		Services		Nmap Output		Ports / Hosts		Topology		Host Details		Scans	
OS	Host												
	pentest-ground	<pre>nmap -p 81 --script http-enum pentest-ground.com Starting Nmap 7.99 (https://nmap.org) at 2026-07-02 21:48 +0300 Nmap scan report for pentest-ground.com (178.79.134.182) Host is up (0.16s latency). rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com PORT STATE SERVICE 81/tcp open hosts2-ns Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds</pre>											

SSL Cipher Strength Scan

Tests SSL/TLS ciphers-Cryptographic Failures.



Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

pentest-ground

nmap -p 443 --script ssl-enum-ciphers pentest-ground.com

Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 21:51 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.16s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE
443/tcp	open	https
ssl-enum-ciphers:		
TLSv1.2:		
ciphers:		
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA (secp256r1) - A		
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 (secp256r1) - A		
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (secp256r1) - A		
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA (secp256r1) - A		
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 (secp256r1) - A		
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (secp256r1) - A		
TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 (secp256r1) - A		
compressors:		
NULL		
cipher preference: client		
TLSv1.3:		
ciphers:		
TLS_AKE_WITH_AES_128_GCM_SHA256 (X25519MLKEM768) - A		
TLS_AKE_WITH_AES_256_GCM_SHA384 (X25519MLKEM768) - A		
TLS_AKE_WITH_CHACHA20_POLY1305_SHA256 (X25519MLKEM768) - A		
cipher preference: client		
_ least strength: A		

Nmap done: 1 IP address (1 host up) scanned in 8.64 seconds

SQL Injection Detection Scan

Detects SQL injection points -Injection Attacks.

The screenshot shows the Zenmap interface with the following details:

- Target:** pentest-ground.com
- Command:** `nmap -p 81 --script http-sql-injection pentest-ground.com`
- Hosts:** pentest-ground.com
- Nmap Output:**

```
nmap -p 81 --script http-sql-injection pentest-ground.com

Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 21:54 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.16s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT      STATE SERVICE
81/tcp    open  hosts2-ns

Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds
```

Service Version Detection Scan

Identifies running services/versions -Insecure Design.

The screenshot shows the Zenmap interface with the following details:

- Target:** pentest-ground.com
- Command:** `nmap -sV -p 81 pentest-ground.com`
- Hosts:** pentest-ground.com
- Nmap Output:**

```
nmap -sV -p 81 pentest-ground.com

Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 21:56 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.16s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT      STATE SERVICE VERSION
81/tcp    open  ssl/http nginx 1.31.2

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.97 seconds
```

Configuration Backup Scan

Finds exposed config/backup files-Security Misconfiguration.

The screenshot shows the Zenmap application window. The 'Target' field is set to 'pentest-ground.com'. The 'Command' field contains 'nmap -p 81 --script http-config-backup pentest-ground.com'. The 'Nmap Output' tab is selected, displaying the following text:

```
nmap -p 81 --script http-config-backup pentest-ground.com

Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 22:01 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.16s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT      STATE SERVICE
81/tcp    open  hosts2-ns

Nmap done: 1 IP address (1 host up) scanned in 0.88 seconds
```

Vulnerability Database Scan

Checks for outdated software with CVEs -Vulnerable Components.

The screenshot shows the Zenmap application window. The 'Target' field is set to 'https://pentest-ground.com:5013'. The 'Command' field contains 'nmap -sV -p 81 --script vuln https://pentest-ground.com:5013'. The 'Nmap Output' tab is selected, displaying the following text:

```
nmap -sV -p 81 --script vuln https://pentest-ground.com:5013

Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 22:17 +0300
Unable to split netmask from target expression: "https://pentest-ground.com:5013"
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 10.41 seconds
```


Authentication Testing Scan

Tests login/authentication endpoints-Authentication Failures

 Zenmap

Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

 pentest-ground

nmap -p 81 --script http-auth pentest-ground.com

Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 22:09 +0300
Nmap scan report for [pentest-ground.com](https://nmap.org) (178.79.134.182)
Host is up (0.23s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE
81/tcp	open	hosts2-ns

Nmap done: 1 IP address (1 host up) scanned in 3.20 seconds

File Upload Exploit Scan

Detects unsafe file upload endpoints -Software/Data Integrity Failures.

 Zenmap

Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services Nmap Output Ports / Hosts Topology Host Details Scans

Service

hosts2-ns

https

nmap -p 81 --script http-fileupload-exploiter pentest-ground.com

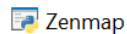
Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 22:14 +0300
Nmap scan report for [pentest-ground.com](https://nmap.org) (178.79.134.182)
Host is up (0.17s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE
81/tcp	open	hosts2-ns

Nmap done: 1 IP address (1 host up) scanned in 0.89 seconds

Error Message Scan

Finds verbose error messages -Logging/Monitoring Failures.



Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services

Service

hosts2-ns

https

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -p 81 --script http-errors pentest-ground.com

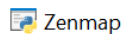
Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 22:15 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.17s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE
81/tcp	open	hosts2-ns

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds

Open Proxy Scan

Detects open proxy misconfigurations -SSRF (Server-Side Request Forgery)



Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services

Service

hosts2-ns

https

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -p 81 --script http-errors pentest-ground.com

Starting Nmap 7.99 (<https://nmap.org>) at 2026-07-02 22:16 +0300
Nmap scan report for pentest-ground.com (178.79.134.182)
Host is up (0.16s latency).
rDNS record for 178.79.134.182: 178-79-134-182.ip.linodeusercontent.com

PORT	STATE	SERVICE
81/tcp	open	hosts2-ns

Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds

OWASP Research Report

1. Broken Access Control

Users gain access to resources or functions beyond their privileges.

For example: An attacker accesses /admin without authorization.

Impact: Unauthorized data exposure, privilege escalation.

Prevention: Role-based access control, least privilege, proper authorization checks.

2. Cryptographic Failures

Sensitive data not properly protected with encryption.

For example: Plain-text passwords stored in a database.

Impact: Data theft, identity compromise.

Prevention: Strong encryption (AES, TLS 1.2/1.3), secure key management, HTTPS enforcement.

3. Injection Attacks

Untrusted input interpreted as commands.

For example: SQL Injection in login forms (' OR '1'='1').

Impact: Unauthorized access, data manipulation, full system compromise.

Prevention: Parameterized queries, prepared statements, strict input validation.

4. Insecure Design

Weaknesses caused by poor application design.

For example: Missing security requirements in business logic.

Impact: Exploitable workflows, bypassed controls.

Prevention: Security-by-design principles, threat modeling, secure architecture reviews.

5. Security Misconfiguration

Improperly configured systems, servers, or applications.

For example: Default passwords left unchanged, exposed config files.

Impact: Easy exploitation by attackers.

Prevention: Harden configurations, remove unused services, regular audits.

6. Vulnerable and Outdated Components

Using software with known vulnerabilities.

For example: Outdated Apache or PHP versions flagged by CVEs.

Impact: Exploitation of known flaws, remote code execution.

Prevention: Patch management, dependency monitoring, regular updates.

7. Authentication Failures

Weak authentication mechanisms allow unauthorized access.

For example: Weak passwords, session hijacking.

Impact: Account takeover, credential stuffing attacks.

Prevention: MFA, strong password policies, secure session management.

8. Software and Data Integrity Failures

Failures in code or updates compromise integrity.

For example: Unsafe file upload endpoints accepting .php shells.

Impact: Remote code execution, malware injection.

Prevention: Validate file types, code signing, secure CI/CD practices.

9. Security Logging and Monitoring Failures

Insufficient logging prevents detection of attacks.

For Example: Verbose error messages leaking database info.

Impact: Delayed incident response, undetected breaches.

Prevention: Centralized logging, alerting systems, incident response planning.

10. Server-Side Request Forgery (SSRF)

Server tricked into making unauthorized requests.

For example: Open proxy misconfiguration allows internal network access.

Impact: Data exposure, cloud service compromise.

Prevention: URL validation, restrict outbound requests, network segmentation.

Security Improvement Proposal

- ✚ Enforce HTTPS/TLS.
- ✚ Strong authentication (MFA, password policies).
- ✚ Input validation (prepared statements).
- ✚ Logging + monitoring with alerts.
- ✚ Regular patching of outdated components

Web Security Checklist

- ✓ Authentication
- ✓ Authorization
- ✓ Input Validation
- ✓ Session Management
- ✓ Encryption
- ✓ Logging
- ✓ Configuration

Reflection

Using Nmap on pentest-ground.com:81, I detected vulnerabilities aligned with OWASP Top 10, including SQL injection, outdated components, and misconfigurations. This reinforced secure coding, patch management, and authentication as essential practices for effective web application security